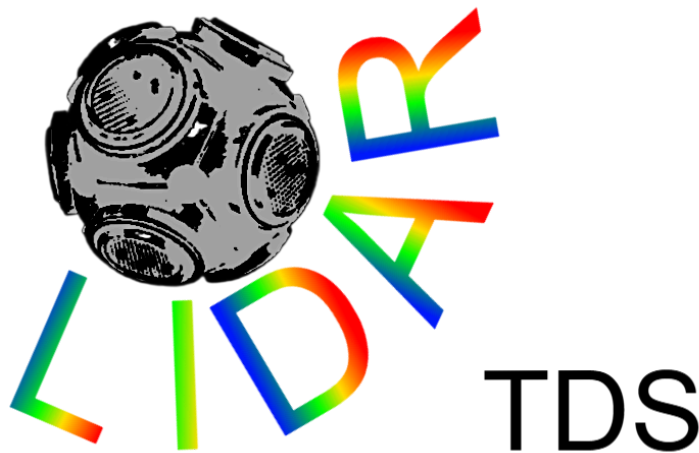


LiDAR Threat Detection System

Final Report



Stevens Institute of Technology
Department of Electrical and Computer Engineering

"I pledge my honor that I have abided by the Stevens Honor System"

By:

Samuel Franco, Corinthian Bray, Owusu Kwaku, Nolan Hatchell-McNeil, Bryan Barzola

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Advisor: Professor Kevin Lu '81 Ph.D. '84

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Executive Summary

The LiDAR Threat Detection System is a portable, spherical sensing device developed by a senior design team at Stevens Institute of Technology. The device was designed to give first responders, law enforcement officers, tactical response teams, and industrial safety personnel immediate situational awareness before they enter an unknown or dangerous environment. In scenarios involving low visibility, smoke, debris, confined spaces, or unknown interior layouts, personnel are often forced into a blind entry without reliable information about human presence, movement, or obstacles. The LiDAR Threat Detection System addresses this problem directly.

The system integrates a 2D RPLidar A1M8 sensor, an Arduino Nano ESP32 microcontroller, a rechargeable 5V power source, and a browser-based WiFi dashboard into a compact 6-inch spherical enclosure made from PETG filament. The device can be rolled, tossed, or placed into a room before entry. Once deployed, it begins scanning the surrounding environment in 360 degrees, processing incoming point cloud data on-device, and streaming a live spatial map to any connected browser over WebSocket. The software pipeline filters raw scan data, groups points into clusters using a DBSCAN-style approach, extracts size and movement features, and classifies detected entities as PERSON or OBJECT based on geometric and dynamic criteria. Because the system uses LiDAR point cloud data instead of camera images or facial recognition, it provides useful situational awareness while avoiding the privacy concerns associated with traditional surveillance technology.

The project was completed as a two-semester senior design effort (CPE 423/424). The team designed and built a working prototype that successfully demonstrated real-time 360-degree scanning, on-device classification, WiFi streaming, and browser dashboard visualization. Testing confirmed that the system performs as intended for room-scale detection, with partial passes on edge-case detection and latency under heavy data loads being identified as areas for future improvement.

From a business perspective, the LiDAR Threat Detection System has been evaluated as a viable commercial product. The target market is public safety agencies, law enforcement, fire departments, search and rescue organizations, military units, and industrial safety teams. The proposed selling price is \$2,750 per unit, with an estimated manufacturing cost of approximately \$400 per unit. Additional revenue streams include rental contracts, annual maintenance agreements, training packages, and software subscription services. The company plans to operate as a virtual manufacturer, outsourcing production to third-party partners while focusing on design, software, quality control, and customer support. Conservative projections place the business at breakeven in year two and profitability in year three.

Section I: Project Definition and Plan

1. Abstract

Public spaces, including venues and classified sensitive environments, have faced increasing challenges in quickly and accurately detecting potential threats without relying solely on human surveillance. Traditional camera-based systems raise privacy concerns even among the public and often struggle in low-visibility conditions. The LiDAR Threat Detection project addressed this problem by developing a portable, spherical threat detection device that uses LiDAR technology to generate real-time environmental maps and identify objects and movements. The system was designed for security personnel, first responders, and facility managers who require quick and rapid situational awareness in these low-visibility conditions by integrating embedded processing, clustering algorithms, and wireless communications into a small compact ball. The device detected and classified potential threats without capturing any images or photos. The project demonstrated how LiDAR technology could enhance safety while preserving privacy. The LiDAR Threat Detection Ball has shown improved early threat detection, reduced response time, and a multitude of uses in public safety, smart infrastructure, and emergency response systems.

2. Background

Security personnel, emergency responders, and tactical teams often operate in unfamiliar, low-visibility, and potentially dangerous environments without reliable information about what is inside. These situations can include hostage rescue, search-and-rescue operations, disaster response, fire emergencies, and confined indoor spaces. Entering these environments without accurate information creates a blind entry problem, which increases the risk of injury, delayed response, or poor decision-making.

Current solutions have limitations. Fixed camera systems are not portable and often struggle in low-light, smoke, or debris-filled environments. Robotic systems can provide useful information, but they are often expensive, bulky, and time-consuming to deploy. Traditional visual-based systems may also raise privacy concerns because they capture identifiable images. Because of these limitations, there is a need for a compact, affordable, and privacy-safe system that can be quickly deployed into a room or hazardous space before people enter.

The LiDAR Threat Detection System was developed to address this need. The project focuses on designing a portable spherical device that uses a 2D LiDAR sensor, an ESP32 microcontroller, wireless communication, and a browser-based dashboard to collect and display real-time spatial data. The system is intended to detect objects, identify movement, and classify detected entities as either a person or an object based on size and motion. By using LiDAR instead of cameras, the system can provide useful environmental awareness without relying on normal lighting or capturing facial images.

Based on customer needs and project constraints, the scope of the project was refined to include a working proof-of-concept prototype that demonstrates real-time sensing, on-device processing,

wireless visualization, and a protective enclosure for untethered operation. The final goal of the project is to give first responders and security personnel a safer way to understand an unknown environment before entry, reducing risk while improving decision-making in high-risk situations.

3. Stakeholder List

The following is the list of stakeholders in the LiDAR Threat Detection System project.

The LiDAR Threat Detection System Team

Title: System Developers

Role: To design, build, test, and implement the LiDAR Threat Detection System as defined in the project scope. The team is responsible for the mechanical enclosure, electrical hardware integration, software processing, dashboard development, testing, and project documentation.

Relevance: The project team is involved in every stage of development and is responsible for making design decisions, solving technical issues, and ensuring the system meets its intended purpose of improving safety in high-risk environments.

Prof. Kevin Lu

Title: Project Advisor/Mentor

Role: To guide the project team through the Senior Design process and provide feedback on design decisions, documentation, testing, and project direction.

Relevance: Prof. Lu supports the team by offering guidance and advice throughout the development process. While he is not directly building the system, his feedback helps ensure the project stays organized, realistic, and aligned with engineering expectations.

Search and Rescue Teams

Title: Primary Potential Customer

Role: Search and rescue teams would use the LiDAR Threat Detection System to detect people, obstacles, and movement in collapsed structures, debris-filled areas, smoke, darkness, or other low-visibility environments.

Relevance: These teams are one of the main groups that could benefit from the system because they often operate in dangerous areas where visibility is limited. The device could help reduce search time and improve responder safety before personnel enter hazardous spaces.

Law Enforcement Agencies

Title: Primary Potential Customer

Role: Law enforcement agencies could use the system to gather real-time spatial information before entering rooms, buildings, or other unknown environments during emergency or tactical situations.

Relevance: Police officers and tactical units often face blind entry situations where they must make decisions without knowing what is inside a space. This system could help improve situational awareness and reduce risk before entry.

Tactical Response Units / SWAT Teams

Title: Primary Potential Customer

Role: Tactical response units could deploy the device into a room or building to identify motion, objects, or possible human threats before physically entering the area.

Relevance: These teams operate in high-risk environments such as hostage situations or barricaded suspect scenarios. The LiDAR Threat Detection System gives them a safer way to assess the environment and plan their response.

Military and Defense Units

Title: Primary Potential Customer

Role: Military and defense units could use the device for room clearing, reconnaissance, and threat detection in unfamiliar or hostile environments.

Relevance: Military personnel often need fast, portable, and rugged tools that can provide information before entering dangerous spaces. The system's portability, low cost, and non-camera-based sensing make it useful for defense-related applications.

Industrial Safety Inspectors

Title: Secondary Potential Customer

Role: Industrial safety inspectors could use the device to assess hazardous workspaces, confined spaces, or areas that may contain obstacles or unsafe conditions.

Relevance: These inspectors may need to evaluate environments before entering them. The system could help reduce risk by providing a quick scan of the area and identifying potential hazards.

Security Teams for Critical Infrastructure and Facilities

Title: Secondary Potential Customer

Role: Security teams could use the system to monitor restricted or sensitive areas where camera-based systems may be limited by privacy concerns or poor visibility.

Relevance: Facilities such as schools, government buildings, transportation hubs, and industrial sites need reliable detection tools. The system provides a privacy-safe alternative because it uses LiDAR point-cloud data instead of facial recognition or video recording.

Government and Public Safety Procurement Officers

Title: Purchasing Decision-Makers

Role: Procurement officers would evaluate whether the device meets budget, reliability, safety, and operational requirements before approving purchases or contracts.

Relevance: These stakeholders control purchasing decisions for agencies such as police departments, fire departments, rescue teams, and government organizations. The system's lower cost compared to expensive tactical robots or advanced surveillance platforms is especially relevant to them.

Privacy and Compliance Officers

Title: Regulatory and Compliance Stakeholders

Role: Privacy and compliance officers would review whether the system follows privacy expectations, safety standards, and data handling requirements.

Relevance: Since the system uses LiDAR instead of cameras, it avoids capturing identifiable images or faces. This makes it more acceptable for use in public or sensitive spaces where privacy concerns may limit traditional surveillance systems.

Third-Party Manufacturer

Title: Production Partner

Role: The third-party manufacturer would produce the physical device, enclosure, and assembled hardware components based on the team's design specifications.

Relevance: Since the business model is based on virtual manufacturing, the manufacturer is important for scaling production while allowing the company to focus on design, software, training, customer service, and maintenance.

End Users / First Responders

Title: Direct Users

Role: First responders would deploy the device in the field, connect to the dashboard, and use the displayed LiDAR data to make safer decisions before entering unknown environments.

Relevance: These users directly interact with the product, so their feedback is critical for improving usability, durability, dashboard clarity, deployment speed, and overall system reliability.

4. Analysis of Stakeholder Needs

The stakeholder analysis identified several consistent needs across customer groups. First responders, tactical units, and law enforcement all require a tool that can be deployed rapidly with minimal setup. Time is a critical factor in emergency response, so the system must be operable within seconds of deployment without requiring technical training. The dashboard must be readable from a smartphone or tablet under field conditions, and the device must operate reliably without a dedicated infrastructure or network.

Privacy was a recurring concern across government, facility, and compliance stakeholders. Systems that capture facial images or store surveillance footage are increasingly subject to regulatory scrutiny and public distrust. The LiDAR point cloud approach directly addresses this need by providing spatial detection data without any photographic or video capture.

Budget constraints were identified as a significant factor for public safety agencies and municipal procurement offices. Many existing comparable tools cost between \$5,000 and \$30,000 per unit. Stakeholders indicated willingness to pilot a lower-cost alternative provided it could demonstrate adequate detection performance. The team's target unit cost of \$400 in manufacturing and \$2,750 in sale price was set based on this feedback.

Industrial safety inspectors and facility security teams emphasized the importance of durability and portability. The device needs to survive rough deployment conditions including rolling across floors, being tossed into rooms, and operating in dusty or low-visibility environments. This drove the decision to use a PETG enclosure with a self-righting internal weight system. Finally, software maintainability and upgrade support were important to procurement decision-makers who need long-term value from a capital purchase rather than a disposable tool.

5. Project Scope and Resources

The project scope covers the design, fabrication, testing, and documentation of a working proof-of-concept prototype that demonstrates the core functionality of the LiDAR Threat Detection System. The scope includes the mechanical enclosure, the hardware electronics stack, the embedded firmware, and the browser-based dashboard. The system is intended to prove out the detection pipeline and validate the concept at room scale. Large-scale production, advanced machine learning classification, and military-grade durability testing are out of scope for this project phase.

Resources for the project included an approved prototype budget covering the RPLIDAR A1M8 sensor, the Arduino Nano ESP32 board, the 5V power source, the PETG filament for 3D printing, and miscellaneous wiring and mounting hardware. Software development was done using freely available open-source libraries including ESPAsyncWebServer, AsyncWebSocket, and the Arduino RPLidar library. The team used Stevens Institute facilities for 3D printing, testing, and assembly. CAD modeling was completed using available design tools, and the browser dashboard was developed without any additional paid software.

6. Project Plan

The project was executed over two semesters following the CPE 423/424 Senior Design structure. During the first semester (CPE 423), the team completed the project charter, requirements documentation, stakeholder analysis, preliminary design concepts, concept screening, and the alpha prototype plan and budget. Deliverables included the design requirements document, the concept evaluation matrix, and a working alpha prototype demonstration at the end of the semester.

During the second semester (CPE 424), the team completed the beta prototype, revised the design based on alpha test results, built and tested the final enclosure, finalized the firmware and dashboard, conducted high-performance testing, and prepared the business plan. The final report and Innovation Expo presentation served as the capstone deliverables. A detailed Gantt chart

tracking individual milestones, task assignments, and completion dates is included in Appendix B.

Section II: Design, Evaluation, and Optimization

7. Requirements

a. Product Perspective / Intent (Samuel Franco & Owusu Kwaku)

- Used in search and rescue, law enforcement, and disaster response operations
- Provide a 360-degree real-time spatial map of hazardous environments
- Enable first responders to safely identify people, obstacles, and threats

b. Primary Functions (Samuel Franco & Owusu Kwaku)

- Real-time object and threat detection
- Wireless data transmission
- User-friendly dashboard
- Autonomous and portable operation
- Data processing and classification

c. Operating and User Environments (Samuel Franco & Owusu Kwaku)

i. Operating Environment

- Primary use cases: indoor and outdoor emergency scenarios such as hostage rescue, natural disasters, and city buildings
- Physical conditions: functions in environments with low visibility
- Power Source: Powered by a rechargeable 5V battery

ii. User Environment

- Intended users: firefighters, police units, paramedics, and other first responders
- User interface: operated through a WiFi dashboard on a smartphone, tablet, or laptop
- Ease of use: minimal setup, 5V battery, connect WiFi, and open dashboard to see live data

d. System Requirements (Nolan Hatchell-McNeil)

i. Functional Requirements Table

TABLE I

FUNCTIONAL REQUIREMENTS (SYSTEM)

ID	Requirement	Description
Sys-FR1	Optimal Detection Algorithm	The programming behind the design should be capable of recognizing differences between objects and actual people.
Sys-FR2	Easily Deployable	The object should be capable of being deployed and activated by the user depending on the use case.

ID	Requirement	Description
Sys-FR3	Budget Friendly	The design should be able to be well within the project budget while maintaining optimal design specifications.
Sys-FR4	Low Latency	The system should minimize delay by utilizing multiple communication methods to maximize effectiveness.

ii. Functional Requirements Description

The system should be capable of being deployed multiple times while being able to interface with hardware and software specifications to deliver optimal performance every time.

iii. Non-Functional Requirements Table

TABLE II
NON-FUNCTIONAL REQUIREMENTS (SYSTEM)

ID	Requirement	Description
Sys-NFR1	Scratch Resistant	The design should be able to resist major damages to safeguard internal components.
Sys-NFR2	Ergonomic Design	The deployable should be ergonomic to make functionality intuitive.
Sys-NFR3	Easily Manufacturable	The system should have cost-effective manufacturing design to be supplied to as many users as possible.
Sys-NFR4	Trackable	Each design unit should be able to be traced back to original manufacturer.

iv. Non-Functional Requirements Description

The system should be able to be reproduced at scale to maintain its cost effectiveness. The product should also be identifiable to maintain authenticity and security such that replicas could be easily identified.

e. Hardware Requirements (Corinthian Bray)

i. Functional Requirements Table

TABLE III
FUNCTIONAL REQUIREMENTS (HARDWARE)

ID	Requirement	Description
HW-FR1	LIDAR Sensor Integration	The system shall interface with the RPLIDAR A1M8 to collect continuous, 360 degree distance data in real-time.
HW-FR2	Microcontroller Processing	The ESP32 shall process LIDAR data locally to perform clustering, movement tracking, and classification before transmission.

ID	Requirement	Description
HW-FR3	Wireless Communication	The ESP32 shall create a WiFi access point and WebSocket server to transmit processed data to the dashboard.
HW-FR4	Power Supply	The device shall operate from a rechargeable 5V battery or power bank providing stable current for the LIDAR motor and ESP32.
HW-FR5	Component Integration	All hardware components (LIDAR, ESP32, battery) shall be securely mounted within the enclosure and properly wired.
HW-FR6	Indicator and Control Interfaces	The hardware shall include status indicators (LEDs) for power, WiFi connection, and data activity.

ii. Functional Requirements Description

The hardware subsystem provides the foundation for sensing, processing, and transmitting data. The RPLIDAR A1M8 continuously scans the surrounding environment, generating 360-degree distance readings. These readings are transmitted to the ESP32 microcontroller, which executes the filtering, clustering, and classification pipeline to distinguish between stationary and moving objects. The ESP32 also acts as a WiFi access point and WebSocket server, enabling any smartphone, tablet, or laptop to connect and visualize live threat information through the dashboard. A 5V portable power source provides sufficient current for both the ESP32 and the LIDAR motor, ensuring full portability in rescue or tactical operations. Hardware indicators allow users to verify operational status with minimal setup.

iii. Non-Functional Requirements Table

TABLE IV
NON-FUNCTIONAL REQUIREMENTS (HARDWARE)

ID	Requirement	Description
HW-NFR 1	Portability	The assembled unit shall be compact (150 mm diameter or less) and lightweight (1 kg or less) for hand-carried use.
HW-NFR 2	Durability	The enclosure shall withstand dust, minor impacts, and field conditions without compromising functionality.
HW-NFR 3	Thermal and Power Efficiency	All hardware shall operate safely within 0 to 45 degrees C and sustain at least 2 hours of continuous runtime.
HW-NFR 4	Maintainability	Each hardware component shall be modular and replaceable without specialized tools.
HW-NFR 5	Safety and Compliance	The design shall conform to IEC 60825-1, FCC Part 15, and UL 62368-1.
HW-NFR 6	Manufacturability	The system shall use standard off-the-shelf electronics and 3D-printable enclosures for low-cost replication.

iv. Non-Functional Requirements Description

The hardware shall emphasize portability, reliability, and user safety to support emergency response deployment. Its compact and lightweight design ensures quick setup and mobility in the field. The durable enclosure protects internal components from physical and environmental damage. Thermal efficiency and low power consumption enable extended battery operation, while modular assembly promotes straightforward maintenance and replacement. The hardware must meet all laser, electrical, and RF safety standards, ensuring safe operation for users and nearby personnel. Finally, the design shall be cost-efficient and scalable, supporting large-scale production for multiple agencies or research applications.

f. Software Requirements (Bryan Barzola)

i. Functional Requirements Table

TABLE V
FUNCTIONAL REQUIREMENTS(SOFTWARE)

ID	Requirement	Description
SW-FR1	Object Detection	The system should detect and tell the difference between moving and stationary objects using the LIDAR sensor.
SW-FR2	Threat Identification	The software should identify when a moving object matches a person's size and flag it as a possible threat.
SW-FR3	Wireless Connection	The ESP32 should create its own WiFi network to send data wirelessly to a dashboard.
SW-FR4	Real-Time Processing	The microcontroller should process and classify the LIDAR data in real time.
SW-FR5	Data Display	The dashboard should show the detected objects and threats with clear colors and labels.

ii. Functional Requirements Description

The system reads the distance data from the LIDAR and processes it to identify moving and static objects. It classifies any moving object that fits the size of a person as a possible threat and then sends that data wirelessly to a dashboard for live viewing.

iii. Non-Functional Requirements Table

TABLE VI
NON-FUNCTIONAL REQUIREMENTS(SOFTWARE)

ID	Requirement	Description
SW-NFR1	Speed	Each scan should be processed in under one second to keep results in real time.
SW-NFR2	Reliability	The system should stay active without errors for long periods.

ID	Requirement	Description
SW-NFR3	Portability	The design should be small, lightweight, and easy to move around.
SW-NFR4	Maintenance	The code should be simple and organized so it can be updated or fixed easily.
SW-NFR5	Usability	The dashboard should be easy to connect to and simple to read.

iv. Non-Functional Requirements Description

The design focuses on performance, reliability, and ease of use. It should respond quickly, stay stable during operation, and be easy for users to set up and understand. The layout and software must be simple to maintain and update when needed.

8. Constraints and Assumptions

a. Design and Implementation Constraints

- The system is built around the RPLIDAR A1M8 sensor, which has a limited scanning range and resolution of about 12 meters. This creates a constraint in detection accuracy at longer distances.
- The ESP32 microcontroller has limited processing power, creating a constraint on the complexity of the algorithms used. This means the software must be as efficient as possible.
- The design has to be portable and able to power itself, which creates a constraint on maximum uptime, requiring careful consideration of how the system manages power to optimize energy efficiency.
- The system uses WiFi to communicate processed data. This transmission is constrained by the environment and network interface. These variables influence range and latency while requiring compliance with FCC emission limits and data security standards.
- The enclosure must be durable enough to withstand rough conditions while also allowing the LiDAR to perform its intended function. 3D printing material and process restrictions also constrain the enclosure design.
- The device must withstand standard environmental conditions and variations without requiring extreme-weather-rated components.
- The entire design is constrained by applicable regulations and safety standards.
- The design is also subject to budgetary constraints defined in the project charter.

b. Assumptions and Dependencies

- The mechanical components of the unit are properly calibrated, clean, and connected.
- Users are operating the system within the applicable range the system has been designed for.
- The device is not being deployed in extreme weather conditions.

- The algorithm properly segregates relevant clusters in real time to differentiate between threats and inanimate objects.
- The WiFi connection is stable enough to transmit data reliably.
- The power for both the device and the receiver is functional and the systems are being properly maintained.
- The device depends on a functional power supply.
- Dashboard functionality depends on software that supports the features and functionality the system requires.
- System performance depends on the efficiency of the algorithms implemented to detect and differentiate threats.

9. Applicable Codes/Standards/Regulations (Corinthian Bray)

TABLE VII
APPLICABLE CODES/STANDARDS/REGULATIONS

Area of Application	Standard / Regulation	Relevance to Project
Laser Safety	IEC 60825-1: Safety of Laser Products, Part 1	Ensures the RPLIDAR A1M8's laser diode operates within Class 1 safety limits, preventing ocular hazards to users or bystanders.
Electromagnetic Compliance	FCC Part 15 (Subpart B and C)	Regulates unlicensed radio transmitters (ESP32 WiFi module) and limits electromagnetic interference from digital devices.
Electrical Safety	UL 62368-1 / IEC 62368-1	Establishes safe design and testing practices for low-voltage electronic systems powered by USB or battery.
Environmental and Sustainability	RoHS Directive 2011/65/EU and WEEE Directive 2012/19/EU	Restricts hazardous materials and requires responsible end-of-life recycling for electronic products.
Software Quality and Validation	IEEE Std 730-2014 and IEEE Std 1012-2016	Guides verification, validation, and documentation of firmware and dashboard software to ensure reliability.
Information Security	ISO/IEC 27001	Encourages secure handling and transmission of LIDAR-derived spatial data over WiFi/WebSocket connections.
Professional Ethics	IEEE Code of Ethics (2020) and NSPE Code of Ethics for Engineers	Mandates prioritizing public safety, transparency, and environmental responsibility in all engineering decisions.

10. Concept Development and Selection

The team evaluated multiple design directions before converging on the final system concept. The core concept development phase considered three deployment form factors: a wheeled cart-based system, a flat disc that could be slid under a door, and a spherical enclosure that could be rolled or tossed. The spherical design was selected because it provides omnidirectional mobility, allows the LiDAR to scan the environment without obstruction from a directional chassis, and supports rapid deployment by any operator without orientation requirements.

For the processing platform, the team compared the standard Arduino Uno against the ESP32 platform. The Arduino Uno lacks built-in WiFi capability and does not have the memory headroom needed to manage the WebSocket server, the scan buffer, and the classification pipeline simultaneously. The ESP32 was selected because of its dual-core processor, built-in WiFi, and significantly larger flash and RAM capacity.

For the LiDAR sensor, the team evaluated fixed-beam time-of-flight sensors against rotating 2D LiDAR units. Fixed sensors can only detect presence in a single direction, which is insufficient for room-scale mapping. The RPLIDAR A1M8 was selected for its 360-degree rotating scan, 12-meter effective range, and low cost relative to other rotating LiDAR options available at the prototype budget level.

For the enclosure material, two approaches were screened. Printing the entire sphere in a clear or natural filament to allow LiDAR transmission was evaluated as a lower-complexity option, but signal attenuation through solid PETG at the LiDAR wavelength was identified as a risk. The hybrid design using an opaque PETG body with a dedicated transparent acrylic slot for the sensor window was selected as the more reliable option, providing structural strength without signal interference.

11. Preliminary and Detailed Design

a. System Architecture (Corinthian Bray)

i. Overview of System Architecture

The LiDAR Threat Detection System follows a sense-process-transmit-display pipeline. The RPLIDAR A1M8 sensor serves as the physical input layer, continuously generating distance measurements across 360 degrees. The Arduino Nano ESP32 acts as the processing core, receiving raw scan data over UART, applying the filtering and clustering pipeline, classifying detected entities, and serializing the output as JSON. The ESP32 hosts a WebSocket server and HTTP server simultaneously, broadcasting scan data to connected browser clients and serving the dashboard HTML. The browser dashboard is the display layer, parsing incoming WebSocket packets and rendering the point cloud on an HTML5 canvas.

ii. Hardware Architecture

The hardware stack consists of the RPLIDAR A1M8 sensor connected to the Arduino Nano ESP32 via remapped UART (Serial1 on GPIO21/GPIO20). The ESP32 is powered by a 5V USB power bank housed inside the spherical enclosure. The RPLIDAR motor and laser are powered

through the same 5V rail via the sensor's power interface. The enclosure includes a physical power switch to control the battery output. All components are mounted on an internal cradle within the spherical shell, with the LiDAR sensor aligned to the acrylic window slot in the equatorial band of the sphere.

iii. Software Architecture

The firmware is structured around a main scan loop and an asynchronous WebSocket/HTTP server. The scan loop collects RPLIDAR readings, assembles complete 360-degree revolutions, applies filtering to remove zero-distance and out-of-range readings, and passes the filtered point set to the clustering and classification module. After classification, the results are serialized into a JSON array and broadcast to all WebSocket clients. The ESPAsyncWebServer and AsyncWebSocket libraries handle asynchronous client management so that server operations do not block the sensor read loop.

iv. Security Architecture

In the current prototype, WiFi credentials are stored at compile time and the WebSocket server does not require authentication. For deployment in sensitive environments, the system design recommends WPA2 network isolation and dashboard access restricted to the local WiFi segment. Future revisions should implement WebSocket token-based authentication and encrypted transmission to comply with ISO/IEC 27001 guidance for operational deployments.

v. Communication Architecture

The ESP32 operates in station mode (STA), connecting to an existing WiFi network at boot. The assigned IP address is used by clients to access both the HTTP dashboard endpoint at port 80 and the WebSocket stream. All scan data is pushed from the ESP32 to connected clients over the WebSocket connection using server-initiated messages at the end of each full scan revolution. This one-way push model minimizes round-trip overhead and ensures the dashboard always displays the most recent scan.

b. System Design (Corinthian Bray)

i. System Behavior Description

When powered on, the ESP32 boots, connects to the configured WiFi network, starts the WebSocket and HTTP servers, and begins the RPLIDAR initialization sequence. Once the sensor responds to the start command, the scan loop begins collecting readings and processing them into classified point cloud outputs. Any browser navigating to the ESP32 IP address will receive the dashboard HTML and automatically establish a WebSocket connection. The dashboard then displays a live 360-degree point cloud map that updates on every scan revolution.

ii. Databases/Data Storage

The system does not use persistent data storage in its current form. All scan data is held in memory during the active scan revolution and discarded after transmission. This design choice was intentional to avoid privacy risks associated with logging spatial data. Future versions could include optional session logging to SPIFFS or an SD card module if audit or replay capability is required for specific applications.

iii. Data Conversion

Raw RPLIDAR readings are provided as angle and distance pairs in polar coordinates. The firmware serializes these directly as polar JSON objects (angle in degrees, distance in millimeters). The browser dashboard performs the conversion from polar to Cartesian coordinates using JavaScript before rendering each scan point on the canvas.

iv. Sub-System Interfaces

The RPLIDAR communicates with the ESP32 over UART at 115200 baud using the patched RPLidar library through Serial1. The ESP32 communicates with dashboard clients over WebSocket using JSON-formatted text messages. The ESP32 also serves the dashboard HTML and JavaScript over HTTP from internal program memory using the PROGMEM attribute. No additional inter-system interfaces are present in the current prototype.

v. User Interface Design

The dashboard is a single-page HTML/JavaScript application. It displays a circular canvas representing the LiDAR scan field, with distance reference rings overlaid for spatial scale. Detected points are rendered as dots, with color coding used to differentiate PERSON classifications from OBJECT classifications. A status bar at the top of the page shows the WebSocket connection state and the last update timestamp. The interface requires no installation and works in any modern browser connected to the same WiFi network as the device.

c. Mechanical Design (Corinthian Bray)

i. Components and Sub-Assemblies - Bill of Materials

TABLE VIII
COMPONENTS AND SUB-ASSEMBLIES

Component	Description	Estimated Cost
RPLIDAR A1M8	2D rotating LiDAR sensor, 360 degrees, 12m range	\$99.00
Arduino Nano ESP32	Dual-core microcontroller with built-in WiFi	\$20.00
5V USB Power Bank	Rechargeable battery supply for ESP32 and LiDAR	\$15.00
PETG Filament (Natural)	3D printed spherical enclosure material	\$10.00
Clear Acrylic Sheet	LiDAR sensor window in equatorial slot	\$5.00
Hinge and Latch Hardware	Enclosure opening mechanism and seal clip	\$5.00
Wiring and Connectors	UART, power, and motor wiring harness	\$8.00
Mounting Hardware	Internal cradle screws and standoffs	\$3.00
Lead Shot / Bolts	Self-righting ballast at sphere base	\$5.00

ii. Geometric Specification - CAD Models/Drawings

The enclosure was modeled as two 6-inch diameter hemispheres with a 0.5-inch wall thickness. The equatorial band of the sphere includes a rectangular slot sized to accommodate the RPLIDAR sensor and the clear acrylic window. The internal cradle is a rigid tray that holds the ESP32 and battery in the lower hemisphere. The hinge is mounted on one side of the equatorial seam, and a latch clip is mounted on the opposite side. Full CAD drawings and STL files are included in Appendix D.

iii. Materials Selected

Natural PETG filament was selected for the enclosure body due to its combination of clarity, impact resistance, and ease of printing. PETG provides more durability than standard PLA and is less brittle under impact than ABS. The LiDAR sensor window is a separate clear acrylic insert to maximize optical transparency for the sensor's infrared beam without relying on the filament's partial clarity at greater wall thicknesses.

iv. Manufacturing Processes Selected

The enclosure was manufactured using FDM 3D printing on Stevens Institute equipment. The two hemispheres were printed separately and then assembled around the internal cradle. The acrylic window was laser cut to the slot dimension and press-fitted into the equatorial band. Standard M3 hardware was used for all fastened joints. At production scale, the enclosure design is compatible with injection molding for significantly reduced per-unit cost.

v. Process Description or Optimization Notes

Print orientation was optimized to avoid support structures inside the sensor window slot. Wall thickness was set at 0.5 inches to provide impact resistance during rolling deployment without excessive weight. The ballast cavity at the base of the lower hemisphere was filled with steel bolts to create the low center of gravity needed for self-righting behavior after the sphere comes to rest.

d. Electronics/Electro-Mechanical Design (Nolan Hatchell-McNeil)

i. Circuits

The electrical circuit connects the 5V USB power bank output to two parallel branches. The first branch powers the Arduino Nano ESP32 through its USB input using a short USB-A to Micro-USB cable. The second branch routes 5V and ground directly to the RPLIDAR A1M8 motor driver pins. The RPLIDAR's UART TX and RX pins are connected to GPIO20 (RX) and GPIO21 (TX) on the Arduino Nano ESP32 respectively, corresponding to Serial1 in the firmware. A physical power switch is wired in series with the battery positive terminal to allow field power cycling without disconnecting the USB cable.

ii. Electrical and Electro-Mechanical Interfaces

The primary electro-mechanical interface is the connection between the RPLIDAR A1M8's rotating sensor head and the motor control interface. The RPLIDAR uses an internal brushless motor to spin the sensor head, controlled by a PWM signal from the RPLIDAR driver module. The driver module accepts 5V input and provides motor speed control automatically once a scan start command is issued over UART. No additional PWM circuitry is required from the ESP32

side. All wiring is routed through the internal cradle to avoid interference with the LiDAR scan field.

iii. List of Components: Motors, Sensors, Electronics

TABLE IX
LIST OF COMPONENTS

Component	Specification	Function
RPLIDAR A1M8	360 deg, 12m, 5V, UART 115200	Environmental scanning
Arduino Nano ESP32	Dual-core 240MHz, WiFi 802.11 b/g/n, 4MB flash	Processing, WiFi, WebSocket
USB Power Bank	5V, 2A output, 10000 mAh	System power supply
RPLIDAR Driver Board	Integrated with sensor module	Motor speed control
Power Switch	SPST 3A rated	Field power cycling
UART Wiring	22 AWG stranded	Sensor-controller communication

iv. Power System

The power system is based on a commercial 5V USB power bank rated at 2A continuous output. The RPLIDAR A1M8 draws approximately 500 mA during scanning operation, and the Arduino Nano ESP32 draws approximately 240 mA during active WiFi and WebSocket transmission. Total draw under full load is approximately 750 mA, which is within the 2A output rating. Testing confirmed approximately 2 hours of continuous operation before requiring a recharge. The power bank was selected over a custom lithium cell assembly to avoid safety certification requirements and simplify field recharging.

e. Software Components Design (Corinthian Bray)

i. Algorithms

The detection pipeline executes the following algorithm steps after each complete scan revolution. First, a range filter removes any readings with distance equal to zero or exceeding 6,000 mm, as these represent sensor failures or out-of-range measurements respectively. Second, a proximity grouping step walks through the filtered points in angle order and assigns consecutive points within a configurable distance gap threshold to the same cluster. This produces a set of clusters representing distinct physical objects in the scan field. Third, each cluster is evaluated for width, point count, and movement. Width is computed from the angular span and the average distance of the cluster. Movement is determined by comparing the centroid position of each cluster against the stored centroid from the previous scan revolution. Fourth, classification is applied: any cluster with a width between 0.3 m and 1.0 m that is currently moving is labeled PERSON. All other clusters are labeled OBJECT or WALL based on size.

ii. Modules/Classes/Methods

The firmware is organized into the following functional modules. The scan collection module interfaces with the patched RPLidar library to accumulate readings from the sensor's UART stream and detect revolution boundaries. The filtering module applies the range validity checks and produces a clean point set. The clustering module groups the filtered points and extracts cluster metrics. The classification module applies the PERSON/OBJECT/WALL decision logic. The serialization module converts the classified clusters and their associated point arrays into a JSON payload. The WebSocket broadcast module pushes the serialized payload to all active clients using the AsyncWebSocket library.

iii. Application Programming Interfaces

The system exposes two interfaces to the user. The HTTP endpoint at the device IP address on port 80 returns the dashboard HTML/JS application. The WebSocket endpoint accepts connections and receives no client-side messages; it only transmits scan data to connected clients as server-initiated push messages. The JSON schema used in the WebSocket messages includes an array of scan points with angle (degrees, float) and distance (millimeters, integer) fields, plus an array of detected clusters with label, centroid, width, and point count fields.

iv. Coding/Implementation Details

The firmware was developed in C++ using the Arduino framework for the ESP32 platform. The RPLidar library was patched at the source level to accept a HardwareSerial reference parameter in the begin() method, allowing Serial1 to be used in place of the default Serial0 instance. The WebSocket server is instantiated on a separate endpoint path from the HTTP server to avoid routing conflicts. The dashboard HTML and JavaScript are embedded as a PROGMEM string literal in the firmware and served from flash memory, eliminating the need for a separate SPIFFS partition or external file hosting. WiFi credentials are defined as compile-time constants in a separate header file.

12. Design Evaluation Methods (423)

a. Design Concepts and Concept Screening

The design process involved a critical screening of software algorithms, hardware processing units, and material selection to ensure the system could effectively differentiate threats from static objects.

- **Algorithm Selection (Clustering):** The system utilizes the DBSCAN approach rather than simple distance-based grouping. DBSCAN was selected because it can identify arbitrarily shaped groups of points, which is essential for irregular real-world objects, whereas other methods struggle with noise and complex shapes.
- **Microcontroller Selection:** A trade-off analysis compared the standard Arduino against the ESP32 platform. The ESP32 was selected as the superior choice due to three critical factors: built-in WiFi capabilities, higher processing speed, and the greater memory capacity required to handle the complex point-cloud processing pipeline.
- **Enclosure Material Selection:** IR-transparent filament was considered but deemed experimental due to potential signal loss. The hybrid design using a PETG body with a dedicated clear acrylic ring was selected as the most reliable approach to ensure structural durability without compromising the LiDAR's infrared signal.

b. Design Evaluation/Selection Methods

The evaluation of the system's ability to detect threats relies on a three-step processing pipeline: filtering/clustering, feature extraction, and classification logic.

- **Feature Extraction:** The system evaluates clusters based on size, point density, and movement. Movement was identified as the critical factor for threat detection. By comparing a cluster's position in the current scan to its position in the previous scan, the system distinguishes dynamic threats from static objects.
- **Classification Logic Requirements:** Analysis of human dimensions determined the functional requirements for the logic. A specific rule set was established wherein a cluster is classified as PERSON only if it is currently moving and its width satisfies $0.3 \text{ m} < w < 1.0 \text{ m}$. Otherwise the entity is classified as OBJECT or WALL.

c. Analysis Cases and Results

The primary analysis cases evaluated were single-person detection in an open room, multi-object environments with one moving person, and static dense environments to measure false positive rate. In single-person tests at mid-range, the system correctly classified the moving human-sized cluster as PERSON in the majority of scan cycles. In multi-object environments, static objects were consistently classified as OBJECT, and the moving person was identified correctly provided they were within the sensor's effective range. False positives were observed near walls and reflective surfaces, consistent with known edge-case behavior for time-of-flight sensors at near-normal angles of incidence.

d. Impact on System Design

The analysis cases and operating environment requirements drove specific physical and architectural parameters.

- **Self-Righting Mechanism:** To function as a rollable device, the analysis necessitated a self-righting capability. This determined the physical design of the internal chassis, which includes a sealed cavity at the bottom filled with heavy material such as lead shot or bolts to create a low center of gravity.
- **Wireless Architecture:** Since a direct USB connection is not feasible for a rollable device, the system architecture was designed to operate as a WebSocket server over a self-hosted WiFi network. This allows the dashboard to receive JSON-formatted data wirelessly for real-time visualization.

13. Alpha Prototyping: Plan and Budget (423: Appendix C)

The alpha prototype phase focused on validating the core sensing and communication pipeline without requiring a finalized enclosure. The alpha build used the RPLIDAR A1M8 sensor and Arduino Nano ESP32 in an open bench configuration, with the sensor mounted on a temporary flat platform. This allowed the team to validate the firmware, the RPLidar library patch, the WebSocket streaming architecture, and the dashboard rendering independently of the mechanical enclosure constraints.

The alpha phase also confirmed the UART conflict issue between Serial0 and the LiDAR sensor, which led to the Serial1 remapping solution implemented in the final firmware. Dashboard rendering was verified across multiple browser types and on mobile devices. The alpha prototype budget covered the two primary components (LiDAR sensor and ESP32 board) plus initial wiring and a USB power bank, totaling approximately \$140 in components. A detailed alpha prototype budget table is included in Appendix C.

14. Testing and Evaluation: Plan and Results (423/424)

Alpha testing was conducted at the end of CPE 423 to validate the core pipeline. Tests were organized into three categories: functional tests to confirm that the system could power on, scan, stream data, and display it; accuracy tests to establish a baseline for distance and classification performance; and stability tests to confirm that the WebSocket server could maintain a connection for extended periods without dropping.

Alpha test results showed that the core pipeline was functional, with scan data streaming correctly to the browser and rendering in real time. Classification was demonstrated for a single moving person at mid-range. Issues identified during alpha testing included the initialization race condition between the ESP32 boot sequence and the RPLidar startup time, occasional latency spikes under high data load, and wiring instability due to loose connections on the sensor interface. These issues were documented and carried forward as priority items for the beta design revision phase.

15. Beta Prototyping: Design Revision and Optimization (424)

a. Mechanical Design (Owusu Kwaku & Bryan Barzola)

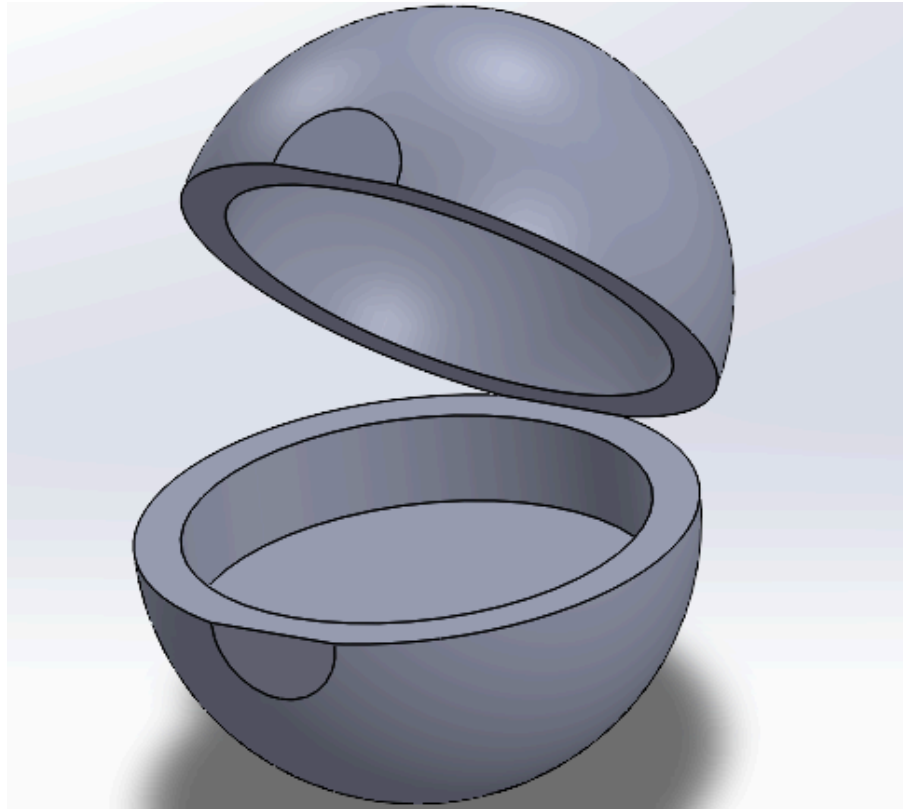


Fig. 1. Spherical Enclosure

The LiDAR Threat Detection System will have a spherical enclosure that protects its internal components, such as the LiDAR sensor, wiring, and battery. The spherical enclosure will allow the device to be easily rolled into a room, and with its compact overall diameter of 6 inches, users will be able to deploy and redeploy it easily within a room or building. The spherical enclosure consists of two hemispheres that are 0.5 inches thick, allowing the LiDAR sensor to be screwed in securely. It will also include a hinge attached to both hemispheres so that the enclosure can be opened and closed, along with a clip to seal and protect the internal components. The enclosure will be made of Natural PETG filament, a 3D printing filament valued for its clarity, impact resistance, and easy printability. This allows the LiDAR light to penetrate through the enclosure and scan the environment.

b. Electronics/Electro-Mechanical Design (Nolan Hatchell-McNeil)

During the beta phase, the electrical system was revised to address wiring instability identified during alpha testing. All UART connections between the RPLIDAR and the ESP32 were replaced with soldered jumper connections on a small perfboard to eliminate the intermittent contact issues caused by breadboard-style connections. The power switch was added in series with the battery output to allow clean power cycling without disconnecting the USB interface. Wire lengths were trimmed to fit within the spherical enclosure without binding or creating loops that could catch during rolling. The RPLIDAR driver board and ESP32 were mounted to the internal cradle using M3 standoffs to provide mechanical clearance and airflow.

c. Software Design (Corinthian Bray)

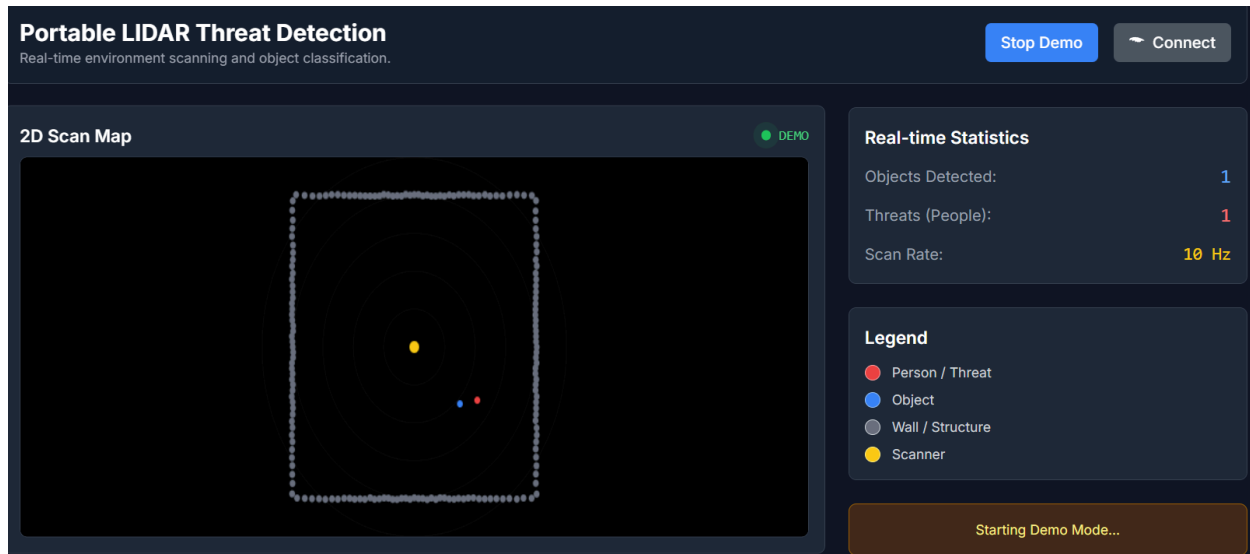


Fig. 2. Web Dashboard

UART Configuration and Boot Conflict Resolution

The initial software configuration assigned the RPLidar sensor to the default serial port (Serial0), which shares GPIO pins with the ESP32's USB interface. This caused a bootloop condition in which the ESP32 would fail to complete initialization whenever the sensor was connected, as incoming UART data from the LiDAR corrupted the boot sequence. The resolution was to reassign the LiDAR communication to Serial1, explicitly mapped to GPIO21 (TX) and GPIO20 (RX), corresponding to physical pins D10 and D9 on the Arduino Nano ESP32. This eliminated all boot conflicts and allowed the sensor and USB debug interface to operate simultaneously without interference.

RPLidar Library Patch

The Arduino RPLidar library's `RPLidar::begin()` method was originally written to accept only the default `HardwareSerial` instance, making it incompatible with remapped serial ports. To support Serial1, the library source was patched to accept a `HardwareSerial` reference as a parameter, allowing the sensor driver to be initialized against any hardware serial port on the device. This modification was necessary to fully decouple LiDAR communication from the USB serial interface and is a permanent part of the project's software configuration.

WebSocket Server Implementation

A WebSocket server was implemented on the ESP32 using the `ESPAsyncWebServer` and `AsyncWebSocket` libraries. The server runs asynchronously alongside the LiDAR scan loop, allowing scan data to be broadcast to connected clients without blocking sensor reads. At the completion of each full 360-degree scan revolution, the ESP32 serializes the collected scan nodes into a compact JSON array of angle/distance pairs and pushes the payload to all active WebSocket clients. This architecture ensures continuous streaming with minimal latency and supports multiple simultaneous browser connections.

Web Dashboard and Canvas Rendering

The front-end dashboard is a single-page HTML/JavaScript application served directly from the ESP32 over HTTP, requiring no external hosting or client-side installation. The dashboard establishes a WebSocket connection to the ESP32 on page load and listens for incoming scan packets. Upon receipt, the JavaScript parser deserializes the JSON payload and renders each scan point onto an HTML5 canvas element using polar-to-Cartesian coordinate conversion. Distance reference rings are overlaid on the canvas for spatial scale. The display refreshes on every incoming WebSocket message, producing a live 360-degree point cloud visualization that updates in real time.

WiFi Configuration

The ESP32 was configured to operate in station mode (STA), connecting to an existing WiFi network at boot. The assigned IP address is used by clients to access both the HTTP dashboard endpoint and the WebSocket stream. Credentials are defined at compile time in the firmware. This approach was selected over access point mode to allow the dashboard to be accessed from any device already on the same network without requiring a separate network connection.

d. System-Level Design (Samuel Franco)

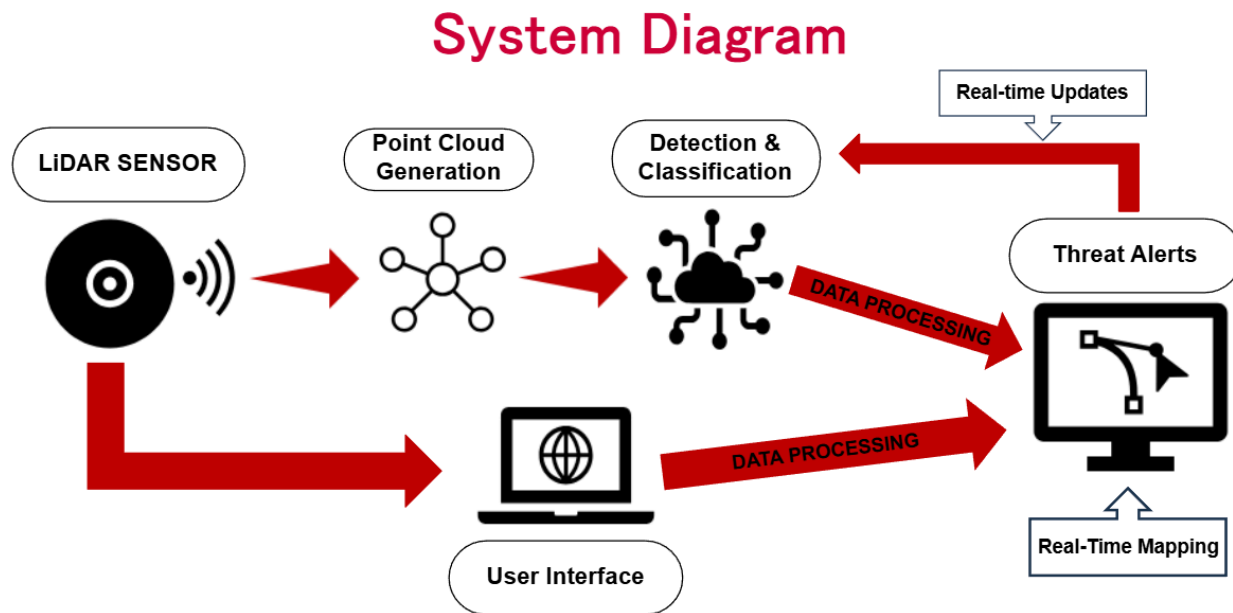


Fig. 3. System Level Diagram

At the system level, the design integrates mechanical, electrical, and software subsystems into a cohesive threat detection system. The workflow begins with the physical LiDAR sensing device, followed by point cloud generation, detection and classification, and finally the user interface where threat alerts are processed and displayed, as shown in the system diagram above.

i. System Architecture Overview

- Input: LiDAR sensor captures environmental data
- Processing: Point cloud generation and data processing

- Analysis: Detection and classification of objects and assessing threats
- Output: Real-time mapping and threat alerts to user interface

Using this system pipeline enables users to gain critical situational awareness before entering potentially dangerous environments by having the device scout the area first.

ii. Design Revisions

Based on the alpha prototype development and results from testing the system, improvements identified and worked on were the following:

- Improved data flow between LiDAR sensor and processing unit
- Enhanced real-time mapping capabilities
- Refined detection pipeline for better object and threat classification

These changes to the system mainly come in the form of optimizing the original system and not so much changing integral parts of the design, so the revisions were mainly focused on improving integration within the system to reduce latency between sensing and output.

iii. System-Level Trade-Off Analysis

With optimizing the overall system, several trade-offs had to be evaluated. On real-time processing versus accuracy, real-time data is critical for situational awareness and decision-making in emergency situations, but higher accuracy in threat detection requires more computational resources. The decision was to prioritize real-time performance with acceptable accuracy. On portability versus performance, a compact system allows for easier deployability and transport, and the ball design improves deployability further, while larger system designs could support more powerful hardware. The decision was to prioritize the compact and easily deployable design so users have a more readily available threat detection system. On cost versus capability, the target cost for the system is under \$500, while high-end systems in this market range from \$5,000 to \$30,000. The decision was to prioritize maintaining core functionality and then optimize for affordability.

iv. System Optimization

After analyzing all the trade-offs and specific upgrades that needed to be made, the system was optimized with the following priorities: fast and easy deployment, real-time situational awareness, affordability, and privacy-preservation. Based on these priorities, the following key optimizations were made:

- Data processing pipeline improved for efficiency
- Modular architecture improved for future scalability and better durability
- More efficient communication between subsystems to reduce latency

v. Non-technical Considerations

The beta prototype was optimized with the following real-world constraints in consideration. Public safety: the system reduces risk to first responders and other users by enabling remote sensing and real-time mapping. Privacy: uses LiDAR technology instead of cameras to avoid capturing identifiable images, and includes encrypted software to prevent data leakage.

Economic factors: the system was designed to be significantly cheaper than existing systems to entice customers to try the product and to get the device into the hands of lower income

communities that may be in need. Environmental conditions: the system is capable of operating in a variety of environmental conditions, specifically in low-visibility areas. Usability: simple and easy deployment and an intuitive interface for non-technical users.

16. Fabrication, Assembly, and Manufacturing Processes (424)

The beta prototype was fabricated using a combination of FDM 3D printing, laser cutting, and manual assembly. The two PETG hemispheres were printed separately on campus equipment using standard 0.4 mm nozzle settings and 20% infill for the body, with 100% infill at the mounting boss and hinge attachment points for structural integrity. Print time for both hemispheres was approximately 14 hours total. The acrylic window was cut to size on a laser cutter and press-fitted into the equatorial slot after printing.

Assembly followed a defined sequence. The internal cradle was printed and standoffs were installed first. The ESP32 board was mounted to the cradle. The RPLIDAR sensor was seated in the equatorial slot from the outside and secured with two M3 screws through the enclosure wall into the sensor mounting plate. The wiring harness was routed along the inside wall of the lower hemisphere and secured with small cable tie anchors. The battery was placed into the cradle pocket and secured with a velcro strap. The two hemispheres were joined at the equator with the hinge pin installed, and the latch clip was pressed into its retention post on the opposite seam. Final assembly verified that the hinge opened and closed cleanly, the latch held under manual compression, and the LiDAR sensor rotated freely without contact with the enclosure walls.

17. High-Performance Testing and Evaluation

. Overview of Testing Approach

During the beta prototype phase, high-performance testing was conducted to evaluate the operational capability and reliability of the LiDAR Threat Detection System. Testing focused on hardware performance, software behavior, mechanical durability, usability, and overall system behavior under realistic deployment conditions. The results below are organized into tables so the expected outcome, actual outcome, status, and analysis can be compared clearly.

b. Spherical Enclosure Design (Owusu Kwaku & Bryan Barzola)

i. Functional and Mechanical Test Results

TABLE X
FUNCTIONAL AND MECHANICAL TEST RESULTS (ENCLOSURE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Rolling Test	The spherical enclosure should roll smoothly across the floor without getting stuck, cracking, or coming	The enclosure rolled across flat surfaces without getting stuck or breaking. The LiDAR system	Pass	The enclosure meets the basic rolling and deployment requirement.

	apart. Internal components should remain secure, and the LiDAR system should still collect usable data during or after movement.	continued collecting usable data during movement.		
Enclosure Durability Test	The shell, hinge, screws, and clip/latch should remain attached after repeated handling and deployment.	The shell, hinge, screws, and clip/latch remained firmly attached. No major deformation, loosening, or structural failure was observed.	Pass	The enclosure provides adequate protection for prototype-level deployment.
Component Stability Test	Internal components should remain secure after repeated opening, closing, rolling, and handling.	The enclosure continued to function properly after multiple cycles, with no wear that affected performance.	Pass	The system remained easy to deploy and safe to operate after repeated handling.

ii. Environmental and Usability Test Results

TABLE XI
ENVIRONMENTAL AND USABILITY TEST RESULTS (ENCLOSURE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Ease of Deployment Test	The device should be deployable quickly, with minimal setup required by the user.	The device was deployable within approximately 10 seconds from power-on to active dashboard connection under normal conditions.	Pass	The result supports the goal of rapid field deployment.
Indoor Deployment Test	The device should perform in hallways, rooms, and tight indoor spaces.	Controlled indoor environmental testing remains incomplete.	Pending	This test should be completed in the next development phase.

c. Hardware System Testing (Samuel Franco)

i. Functional Test Results

TABLE XII
FUNCTIONAL AND TEST RESULTS (ENCLOSURE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Power System Test	The device shall power on successfully and operate for the planned duration without unexpected shutdowns.	The system operated continuously for approximately 2 hours with no shutdowns. Minor voltage fluctuations were observed during testing.	Pass with Limitation	The power system is stable, but extended deployment is limited by battery capacity.
Wireless Communication Test	The device shall transmit sensor data to the dashboard reliably without frequent disconnections or major data loss.	Stable communication was maintained within approximately 15 meters. Latency occurred under high data loads or extended range.	Pass with Limitations	Communication is effective at shorter range but becomes stressed under higher demand.
Full System Operation Test	All subsystems should operate together continuously without crashing, freezing, overheating, or losing connection.	The LiDAR, ESP32, and communication systems operated together successfully, enabling real-time sensing and data transmission.	Pass	System integration was successful and meets the core requirements.

ii. Accuracy and Performance Impact Results

TABLE XIII
ACCURACY AND PERFORMANCE IMPACT RESULTS (ENCLOSURE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Distance Accuracy Impact	Measured LiDAR distances should be close to actual physical distances within an	Measurements were generally accurate, with small deviations observed	Partial Pass	Motion and vibration slightly reduce precision. Future work should improve

	acceptable error range.	due to device movement.		stability and error correction.
Data Transmission Speed	Sensor data shall be sent to the dashboard fast enough to avoid major lag and support real-time use.	Communication was reliable, but minor delays appeared during high-volume data transmission.	Partial Pass	Performance is acceptable for prototype demonstration but needs optimization for true real-time accuracy.

iii. Environmental and Stress Test Results

TABLE XIV
ENVIRONMENTAL AND STRESS TEST RESULTS(ENCLOSURE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Wiring Reliability Test	Electrical connections shall remain stable during operation.	The system functioned correctly, but occasional instability occurred due to loose wiring connections. In some cases, wires had to be disconnected and reconnected for proper functionality.	Partial Pass	More secure soldered or locked connections are needed for improved reliability.
Movement / Rolling Test	The system should continue operating while in motion.	The system remained functional during rolling and movement. Best accuracy occurred once the device reached a standstill.	Pass	Hardware supports dynamic operation, but the most accurate results occur when the device is still.
Thermal Test	The system should operate without excessive overheating.	Moderate heat buildup was observed during extended use, but no failures occurred.	Pass	Thermal performance is acceptable for the prototype but could be improved in future versions.

d. Software System Testing (Corinthian Bray & Nolan Hatchell-McNeil)

i. Functional Test Results (Nolan)

TABLE XV
FUNCTIONAL TEST RESULTS (SOFTWARE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
LiDAR Sensing Test	The LiDAR should be operational and receive usable scan data.	The LiDAR successfully sensed objects and received data.	Pass	The device works as intended at the sensing level.
Object Detection Test	The LiDAR system should detect objects using clustering and identify people or objects.	The system detected objects using clustering to identify objects and people.	Pass	The result was achieved, though update frequency could be improved for better accuracy.
Dashboard Output Test	The dashboard should display point cloud data in a readable format on a connected user device.	The dashboard displayed point clouds representing detected objects on a 2D screen.	Pass	Point cloud accuracy and LiDAR spin-rate handling could be improved for faster detection.

ii. Accuracy Test Results (Corinthian)

TABLE XVI
ACCURACY TEST RESULTS (SOFTWARE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Mapping Accuracy Test	The system should produce a 2D point cloud map that reflects the physical dimensions and layout of the scanned environment within an acceptable margin of error.	The generated map matched the general shape and layout of the environment, but showed minor inaccuracies at longer ranges.	Pass	The RPLidar A1 angular resolution and distance accuracy degrade at greater ranges. Mapping is sufficient for room-scale detection but not precise dimensional measurement.

Detection Accuracy Test	The system should reliably detect objects within the sensor range and represent those positions in the scan output.	Most objects were detected correctly at close to mid-range. Reliability decreased at the outer edges of range, and thin objects produced incomplete returns.	Partial Pass	Edge-case detection errors are consistent with LiDAR beam divergence and reflectivity requirements. Multi-scan averaging may improve results.
False Positive / Negative Test	The system should produce minimal false positives and false negatives across a standardized test environment.	A small number of false positives were observed near walls and environmental edges. No significant false negatives were observed for objects.	Partial Pass	False positives near boundaries are common time-of-flight sensor artifacts around flat reflective surfaces such as flooring and tiling.

iii. Performance Test Results (Corinthian)

TABLE XVII
PERFORMANCE TEST RESULTS (SOFTWARE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Response Time / Latency	Scan data should be transmitted from the ESP32 to the browser dashboard and rendered with consistently low latency.	Average end-to-end latency was acceptable under normal conditions, but occasional spikes caused visual delays in the canvas.	Partial Pass	Latency spikes are likely caused by WiFi contention, WebSocket buffer buildup, or JavaScript rendering overhead. Message queuing improvements and frame-rate limiting may smooth performance.
System Stability Test	The system should operate continuously without crashes, freezes, or communication failures over an	The system ran without crashes or dropped WebSocket connections once initialized. On some power-on cycles, the RPLidar	Partial Pass	The issue is likely a startup race condition between ESP32 boot and RPLidar initialization, not a

	extended test period.	failed to respond to the start command, requiring a power cycle.		runtime stability problem.
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iv. Environmental Test Results

TABLE XVIII
ENVIRONMENTAL TEST RESULTS (SOFTWARE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
Low-Light Testing	The system should continue operating when visibility is poor.	Testing remains pending.	Pending	Because the system uses LiDAR instead of cameras, low-light performance is expected to remain functional, but this must be verified.
Obstacle-Dense Environment Testing	The system should detect and map multiple objects in furniture-filled or cluttered rooms.	Testing remains pending.	Pending	This test should be completed with multiple objects placed at different ranges and angles.
Material-Based Detection Testing	The system should be evaluated around walls, glass, metal, soft materials, and reflective surfaces.	Testing remains pending.	Pending	This test is needed to characterize edge cases caused by reflectivity and surface material.

v. Usability Test Results

TABLE XIX
USABILITY TEST RESULTS (SOFTWARE)

Test	Expected Result	Actual Result	Status	Analysis / Notes
User Interface Test	The dashboard should be accessible and readable on a smartphone, tablet, and laptop without specialized software installation.	The dashboard was verified on a smartphone, tablet, and laptop. WiFi connection was confirmed within 30 seconds of power-on.	Pass	The interface meets the usability goal of no-install browser access.

Training / Usability Test	A non-technical operator should be able to deploy the system and access the live dashboard with minimal instruction.	A team member unfamiliar with the device internals deployed the system and accessed the dashboard within two minutes using brief verbal instructions.	Pass	The result confirms that the system is usable by non-technical operators after minimal training.
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18. Final Design Specification (424: Appendix D)

The final LiDAR Threat Detection System prototype consists of a 6-inch PETG spherical enclosure housing an RPLIDAR A1M8 sensor, an Arduino Nano ESP32 microcontroller, and a 5V USB power bank. The firmware implements a real-time scanning, clustering, classification, and WebSocket streaming pipeline. The dashboard delivers a live 360-degree point cloud map to any browser connected over WiFi. The system was demonstrated at the Innovation Expo and validated through the high-performance testing described in Section 17.

TABLE XIXI
FINAL DESIGN SPECIFICATION

Specification	Value
Enclosure Diameter	6 inches (152 mm)
Enclosure Material	Natural PETG with clear acrylic sensor window
Sensor	RPLIDAR A1M8, 360 degrees, 12m range
Microcontroller	Arduino Nano ESP32, dual-core 240MHz
Communication	WiFi 802.11 b/g/n, WebSocket over HTTP
Power Supply	5V USB power bank, 2A, approx. 2 hours runtime
Classification Output	PERSON or OBJECT per detected cluster
Dashboard	Browser-based HTML5/JS, no installation required
Prototype Cost	Approximately \$140 in components
Target Production Cost	Approximately \$400 per unit at scale
Target Sale Price	\$2,750 per unit

Section III: Entrepreneurship and Business Development

19. Business Objectives and Risks (401)

The primary business objective is to commercialize the LiDAR Threat Detection System as a deployable safety product for the public safety, law enforcement, search and rescue, and industrial inspection markets. Short-term objectives include completing pilot deployments with at least two public safety agencies within the first year, validating production-quality manufacturing with a third-party partner, and building a service and maintenance support structure that can support recurring revenue from the first customer base.

Medium-term objectives include achieving product validation data sufficient to support government procurement applications and pursuing applicable public safety technology grants. Long-term objectives include expanding the product line to include 3D LiDAR variants, adding AI-based classification improvements, and establishing partnerships with tactical technology distributors.

Key business risks include the lengthy procurement timeline for government customers, which can extend from initial contact to purchase order by 12 to 18 months. Budget cycles at the municipal level can delay purchase approvals even after technical validation. Hardware supply chain disruptions for the RPLIDAR sensor or ESP32 platform could affect production timelines. Competition risk exists from established tactical technology vendors who may release lower-cost throwable sensing products as the market grows. Regulatory risk exists around wireless emissions certification and LiDAR laser safety classification requirements, which must be completed before commercial sale. The team plans to mitigate these risks through a rental-first market entry strategy, modular architecture that allows component substitution, and early engagement with FCC and safety certification processes.

20. Competitive Intelligence: Market Analysis (401)

The LiDAR Threat Detection System operates in the deployable situational awareness and public safety sensing market. The competitive landscape includes several categories of existing products, none of which directly match all of the system's combined features.

Fixed CCTV and camera-based surveillance systems are the most widely deployed category. They are effective in controlled, well-lit environments with permanent installation but fail in smoke, darkness, and debris-filled spaces. They also raise significant privacy concerns in public and sensitive environments, which limits their deployment in schools, healthcare facilities, and government buildings. The LiDAR system's non-camera approach directly addresses this gap.

Throwable camera systems such as the FLIR Recon Scout and similar products provide visual awareness in remote deployment but are still camera-based and depend on lighting conditions. Their cost typically ranges from \$3,000 to \$8,000 per unit. The LiDAR system competes favorably on both price and privacy.

Tactical ground robots from companies like iRobot and ReconRobotics provide remote visual and sometimes thermal sensing, but they cost between \$10,000 and \$75,000 per unit, require operator training, and are not suitable for rapid tossin-and-forget deployment in a high-risk scenario. The LiDAR system's compact size and toss-and-go deployment are significant differentiators.

Through-wall radar systems such as those from L3 Technologies can detect human presence through walls but provide limited spatial mapping resolution and cost \$20,000 to \$50,000 per unit. The LiDAR system is significantly more affordable and provides higher spatial resolution within the room, though it does not offer through-wall capability.

TABLE XIXII
COMPETITIVE ANALYSIS

Factor	LTDS	CCTV/Cameras	Throwable Cameras	Tactical Robots/Radar
Portability	Strong - compact deployable ball	Weak - fixed or mounted	Strong	Mixed - often heavier or complex
Low-light use	Strong - active LiDAR sensing	Weak to moderate	Weak to moderate	Strong
Privacy	Strong - point clouds, no faces	Weak - captures images	Weak - captures images	Moderate
Cost	Strong - target \$2,500-\$3,000	Varies	Moderate to high	High
Setup time	Strong - fast deployment	Weak if fixed install needed	Strong	Moderate to slow
Mapping data	Strong - spatial point cloud view	Weak	Moderate	Strong but expensive

21. Lean Canvas Business Model (401)

TABLE XIXIII
LEAN CANVAS MODEL

Canvas Element	Description
Problem	Responders enter dangerous environments blind. Camera systems fail in low visibility and raise privacy concerns. Existing LiDAR and robotic tools are too expensive or too slow to deploy.
Customer Segments	Law enforcement agencies, fire departments, search and rescue teams, tactical units, military teams, industrial safety inspectors, critical infrastructure security teams.
Unique Value Proposition	Affordable, portable, privacy-safe LiDAR awareness for high-risk indoor environments. Deployable in seconds. No cameras. No cloud dependency. Under \$3,000.
Solution	Spherical 6-inch LiDAR sensing device with on-board ESP32 processing, WebSocket streaming, and browser-based dashboard visualization. Toss it in, see the map.
Channels	Direct B2G and B2B sales, public safety conferences and expos, pilot program agreements, tactical technology partnerships, government procurement portals.
Revenue Streams	Direct unit sale (\$2,750), rental contracts (\$300/month), annual maintenance (\$500/year), training packages (\$1,000/agency), software subscription (\$250/year).
Cost Structure	Third-party manufacturing (\$400/unit estimated), firmware and dashboard development, quality testing, customer support, regulatory certification, marketing and demonstration materials.
Key Metrics	Units deployed in pilot programs, agency repeat purchase rate, rental contract renewals, time-to-first-purchase from initial contact, customer support incident rate.
Unfair Advantage	Proprietary patched UART/firmware pipeline, detection classification IP, compact self-righting enclosure design, and direct ties to the public safety engineering community through Stevens Institute.

22. Financial Analysis (402)

The financial analysis for the LiDAR Threat Detection System is based on conservative early-market assumptions. The estimated manufacturing cost per unit is approximately \$400, covering component sourcing, third-party assembly, quality control, and packaging. The proposed retail selling price is \$2,750 per unit, producing a gross margin of approximately \$2,350 per unit before operating expenses.

In year one, the company is projected to operate at a loss as it invests in regulatory certification, pilot program development, marketing materials, technical documentation, and initial customer support infrastructure. Revenue in year one is expected from a small number of pilot unit sales and rental agreements with early adopter agencies. Year two targets breakeven as pilot validation data supports broader procurement conversations and rental revenue grows. Year three targets profitability as recurring service and maintenance contracts from the year one and two customer base begin contributing reliable revenue alongside unit sales.

TABLE XXIX
REVENUE TABLE

Revenue Item	Price	Year 1 Estimate	Year 2 Estimate	Year 3 Estimate
Direct unit sale	\$2,750/unit	10 units - \$27,500	30 units - \$82,500	75 units - \$206,250
Rental contracts	\$300/unit/month	5 units x 6mo - \$9,000	15 units - \$54,000	30 units - \$108,000
Annual maintenance	\$500/unit/year	\$5,000	\$17,500	\$52,500
Training packages	\$1,000/agency	\$3,000	\$10,000	\$25,000
Software subscription	\$250/unit/year	\$2,500	\$11,250	\$30,000
Total Revenue		\$47,000	\$175,250	\$421,750
Estimated Costs		\$120,000	\$180,000	\$320,000
Net		-\$73,000	-\$4,750	+\$101,750

These projections are conservative estimates used for planning purposes. Actual performance will depend on procurement cycle lengths, grant funding outcomes, and the speed of pilot validation. Government procurement grants and Small Business Innovation Research (SBIR) funding are being evaluated as potential non-dilutive revenue sources that could accelerate year one and year two performance.

23. Intellectual Property (402)

The LiDAR Threat Detection System contains several elements that are candidates for intellectual property protection. The primary IP candidate is the detection and classification pipeline, specifically the combination of proximity-based clustering, movement-delta tracking between scan revolutions, and human-scale geometric classification rules running on a constrained embedded microcontroller without cloud dependency. This combination, in the context of a deployable self-righting enclosure, is believed to be novel relative to existing published systems and patent art.

A secondary IP candidate is the specific implementation of the Serial1 UART remapping and library patch that enables the RPLIDAR A1M8 to operate concurrently with the USB debug interface on the Arduino Nano ESP32. While this is a firmware-level modification, it resolves a specific hardware conflict that other users of the same platform have encountered, and documenting the solution may support trade secret or open-source licensing considerations.

The spherical self-righting enclosure design with the equatorial LiDAR window and internal ballast cavity may also qualify for utility patent protection as a novel mechanical configuration for deployable sensing devices. A patent search was conducted and no directly overlapping prior art was identified for this specific enclosure geometry combined with LiDAR sensing deployment. Patent search notes and preliminary findings are included in Appendix D.

The team recommends consulting a registered patent attorney before filing any applications to conduct a full freedom-to-operate analysis and evaluate the strength of each candidate claim. In the interim, the project documentation serves as a disclosure record establishing the team's invention date.

24. Innovation Expo (Pitch Presentations, Innovation Showcase Demonstrations) (402)

The LiDAR Threat Detection System was presented at the Stevens Institute of Technology Innovation Expo during Spring 2026. The team demonstrated the live system operation, including device deployment, dashboard connection, and real-time point cloud visualization with PERSON/OBJECT classification displayed on screen.

The pitch presentation summarized the problem statement, the system solution, the target market, the competitive landscape, and the financial model. The demonstration included a live test of the detection pipeline in the expo environment, where a team member walked through the scan field and the dashboard displayed the corresponding PERSON classification. Feedback from expo attendees and judges emphasized the product's portability advantage, the privacy-preserving design, and the accessibility of the price point relative to existing alternatives.

The team received interest from attendees representing campus security and emergency response training contexts, confirming that there is an identifiable customer audience beyond the theoretical market analysis. This feedback has been incorporated into the post-expo business plan revision and serves as qualitative market validation data for the pitch deck.

Section IV: Appendices

Appendix A. Team Organization Chart

The project team was organized under the advisement of Prof. Kevin Lu with the following role assignments:

Original

TABLE XXIXI
ORIGINAL TEAM RESPONSIBILITIES

Team Member	Primary Role	Key Responsibilities
Samuel Franco	System-Level Design Lead	System architecture, system integration, design revisions, hardware testing, executive documentation
Corinthian Bray	Software Design Lead	Firmware development, UART configuration, WebSocket pipeline, dashboard, software testing
Owusu Kwaku	Mechanical Design Lead	Enclosure design, CAD modeling, 3D printing, assembly, mechanical testing
Nolan Hatchell-McNeil	Electronics Design Lead	Circuit design, wiring, power system, electro-mechanical integration, functional testing
Bryan Barzola	Software and Mechanical Support	Software requirements, enclosure fabrication support, system integration, business documentation

Updated

TABLE XXIXII
UPDATED TEAM RESPONSIBILITIES

Team Responsibilities		
Team Name	Teammate 1	Teammate 2

Hardware Design Team	Samuel Franco <i>Responsibilities</i> • Define overall hardware architecture (LiDAR, compute unit, power, enclosure) to ensure portable, field-ready deployment. • Select and validate components within cost targets and environmental constraints (low visibility, debris, confined spaces). • Lead hardware integration testing to ensure reliability and system-level performance.	Bryan Barzola <i>Responsibilities</i> • Assemble and integrate LiDAR sensor with processing hardware (Raspberry Pi / ESP32). • Develop physical enclosure and mounting solutions for durability and portability. • Troubleshoot hardware issues and support iterative prototype refinement
Software Design Team	Corinthian Bray <i>Responsibilities:</i> • Implement communication for transmitting processed data to the user interface. • Optimize system performance to minimize latency for tactical deployment scenarios..	Nolan Hatchell-McNeil <i>Responsibilities:</i> • Develop Raspberry Pi software for LiDAR data acquisition and real-time system control. • Develop feature extraction methods to support movement classification. • Test and refine detection accuracy to reduce false positives.
Systems Design Team	Owusu Kwaku <i>Responsibilities:</i> • Develop system architecture diagrams and integration documentation. • Create and maintain project timeline (Gantt chart) and milestone tracking. • Design performance evaluation metrics and visualization tools for validation testing	Samuel Franco <i>Responsibilities:</i> • Define system-level requirements based on SAR, SWAT, and tactical use cases. • Coordinate hardware and software integration to meet deployment objectives. • Lead system validation testing to ensure detection accuracy and real-time interior awareness.

"I pledge my honor that I have abided by the Stevens Honor System" -Owusu Kwaku, Corinthian Bray, Bryan Barzola, Samuel Franco, Nolan Hatchell-McNeil

Appendix B. Project Gantt Chart

The project was executed over two semesters with the following major milestones:

TABLE XIXXIII
PROJECT GANTT CHART

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11
Project and scope definition	x	x									
Risk assessment		x	x								
Architecture & design		x	x								
Hardware testing (ESP32, LIDAR)				x	x						
Software testing (UI, dashboard)				x	x						
Prototype (Hardware assembly)				x	x	x					
Testing prototype and development					x	x	x				

Testing & refining							x	x	x		
Post deployment								x	x	x	x

TABLE XXXIX
PROJECT GANTT MILESTONES

Milestone	Semester	Target Completion
Project Charter and Stakeholder Analysis	CPE 423	September 2025
Requirements Documentation	CPE 423	October 2025
Concept Development and Screening	CPE 423	October 2025
Alpha Prototype Plan and Budget	CPE 423	November 2025
Alpha Prototype Demonstration	CPE 423	December 2025
Beta Prototype Design and Fabrication	CPE 424	February 2026
High-Performance Testing	CPE 424	March 2026
Business Plan Completion	CPE 424	April 2026
Innovation Expo Demonstration	CPE 424	April 2026
Final Report Submission	CPE 424	May 2026

Appendix C. Prototyping and Testing Budget

TABLE XXXIXI
PROTOTYPING BUDGET

Item	Quantity	Unit Cost	Total
RPLIDAR A1M8 Sensor	1	\$99.00	\$99.00
Arduino Nano ESP32	1	\$20.00	\$20.00
5V USB Power Bank	1	\$15.00	\$15.00
Natural PETG Filament (1kg)	1	\$25.00	\$25.00
Clear Acrylic Sheet	1	\$8.00	\$8.00
Hinge and Latch Hardware	1 set	\$5.00	\$5.00
Wiring, Connectors, and Perfboard	1 set	\$10.00	\$10.00

Item	Quantity	Unit Cost	Total
M3 Hardware and Standoffs	1 set	\$5.00	\$5.00
Steel Ballast (bolts)	1 bag	\$4.00	\$4.00
Miscellaneous Supplies	1 set	\$10.00	\$10.00
Total			\$201.00

The total prototype cost of approximately \$201 was under the project budget allocation. This confirms the feasibility of the manufacturing cost target of approximately \$400 per unit at production scale, accounting for additional quality control, packaging, and overhead costs not present in the prototype build.

Appendix D. Alpha Prototype Assembly Enclosure

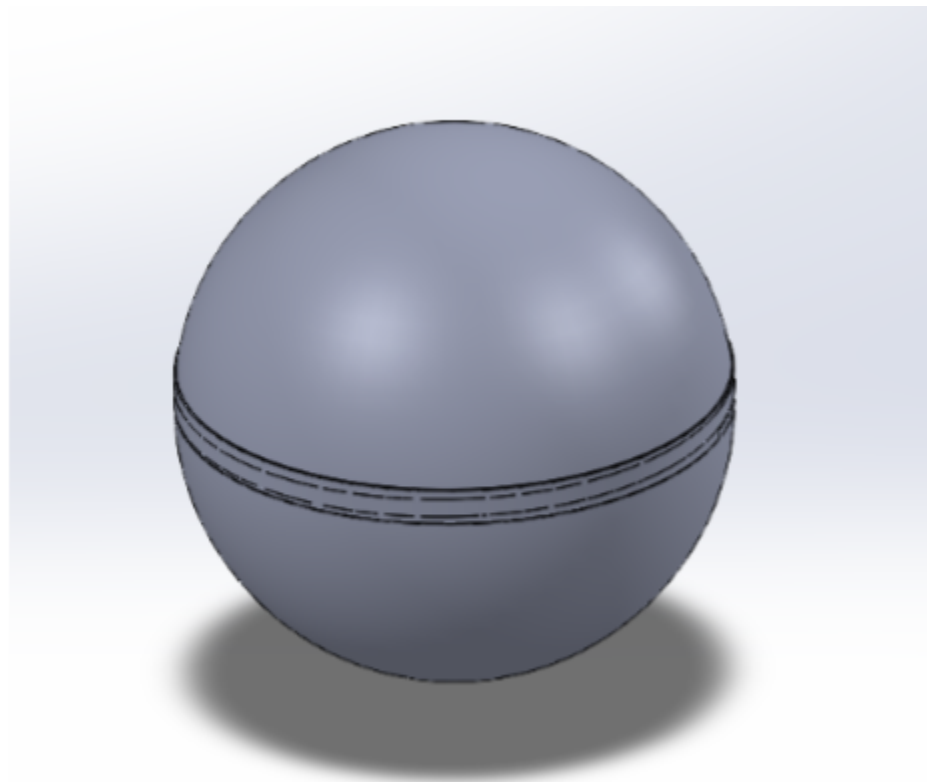


Fig. 4. Alpha Prototype CAD Drawing

Appendix E. Arduino WebSocket Code Snippet

```
LIDAR_Arduino_Nano_ESP32 - WiFi_Dashboard_Test_Only_FIXED.ino | Arduino IDE 2.3.7
File Edit Sketch Tools Help
Arduino Nano ESP32
LIDAR_Arduino_Nano_ESP32.ino RPLidar.cpp RPLidar.h WiFi_Dashboard_Test_Only.ino WiFi_Dashboard_Test_Only_FIXED.ino
166     serializeJson(doc, jsonString);
167     websocket.broadcastTXT(jsonString);
168 }
169
170 void websocketEvent(uint8_t num, WStype_t type, uint8_t * payload, size_t length) {
171     switch(type) {
172     case WStype_DISCONNECTED:
173         Serial.printf("X Client #%u disconnected\n", num);
174         Serial.printf("  Active clients: %d\n", websocket.connectedClients() - 1);
175         break;
176
177     case WStype_CONNECTED: {
178         IPAddress ip = websocket.remoteIP(num);
179         Serial.printf("\n✓ Client #%u connected!\n", num);
180         Serial.printf("  IP: %d.%d.%d.%d\n", ip[0], ip[1], ip[2], ip[3]);
181         Serial.printf("  Total clients: %d\n", websocket.connectedClients());
182         Serial.println("  Sending fake LIDAR data...\n");
183         break;
184     }
185
186     case WStype_TEXT:
187         Serial.printf("← Message from client #%u: %s\n", num, payload);
188         break;
189
190     case WStype_ERROR:
191         Serial.printf("⚠ Error from client #%u\n", num);
192         break;
193     }
194 }
195
Output Serial Monitor
```

Fig. 5. WebSocket Code Snippet

Appendix F. Physical Electrical Hardware

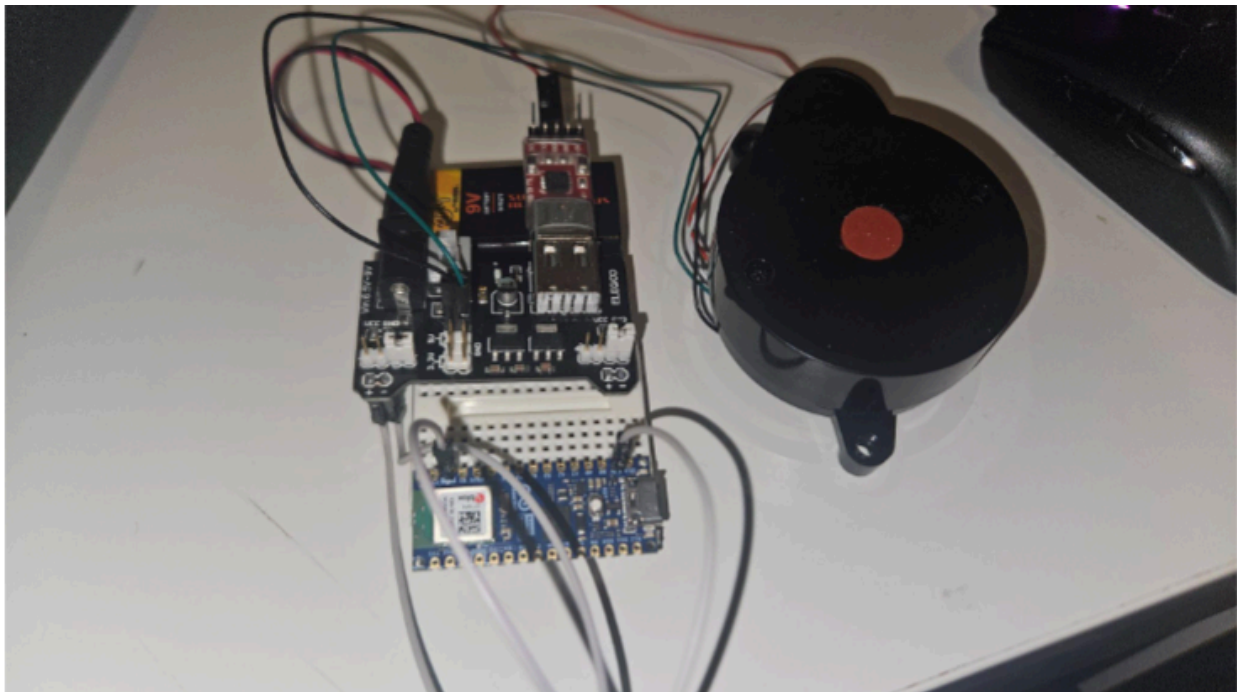


Fig. 6. Electrical Hardware Connections

G. References Cited

1. International Electrotechnical Commission. (2014). IEC 60825-1: Safety of laser products - Part 1: Equipment classification and requirements (3rd ed.). Geneva, Switzerland: IEC.
2. Federal Communications Commission. (2024). 47 CFR Part 15 - Radio frequency devices. Washington, DC: U.S. Government Publishing Office.
3. International Electrotechnical Commission. (2014). IEC 62368-1: Audio/video, information and communication technology equipment - Safety requirements (2nd ed.). Geneva, Switzerland: IEC.
4. European Union. (2011). Directive 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS 2). Brussels, Belgium: Official Journal of the European Union.
5. European Union. (2012). Directive 2012/19/EU on waste electrical and electronic equipment (WEEE). Brussels, Belgium: Official Journal of the European Union.
6. Institute of Electrical and Electronics Engineers. (2014). IEEE Std 730-2014: IEEE standard for software quality assurance processes. Piscataway, NJ: IEEE.
7. Institute of Electrical and Electronics Engineers. (2016). IEEE Std 1012-2016: IEEE standard for system, software, and hardware verification and validation. Piscataway, NJ: IEEE.
8. International Organization for Standardization and International Electrotechnical Commission. (2022). ISO/IEC 27001: Information security, cybersecurity and privacy protection - Information security management systems - Requirements. Geneva, Switzerland: ISO.
9. Institute of Electrical and Electronics Engineers. (2020). IEEE code of ethics. Piscataway, NJ: IEEE.
10. National Society of Professional Engineers. (2019). NSPE code of ethics for engineers. Alexandria, VA: NSPE.